How a Service-Based Company Works on BI Projects

Mahendra Singh

What Is a Service-Based Company?

1. A service-based company focuses on providing expertise and services, not tangible products.
2. Examples include IT consulting, digital marketing, analytics, and software development

Understanding the complete lifecycle of client projects in IT consulting and service-based firms — from initial opportunity identification through successful delivery and long-term partnership building.

01

Introduction & Ecosystem

Understanding service models and partnerships

02

Sales & Acquisition

Identifying opportunities and winning projects

03

Project Initiation

POC validation and formal kick-off

04

Discovery & Planning

Requirements gathering and strategic planning

05

Execution & Delivery

Implementation, validation, and handover

06

Ongoing Partnership

Support and relationship management

Partner Ecosystem

1. Service companies often collaborate with partner firms or client networks.
2. Partners refer projects or collaborate on delivery efforts.
3. Helps maintain a continuous flow of new projects.

Service-based companies rarely work in isolation. They build strategic relationships with **partner firms, technology vendors, and client organizations** to expand their reach and capabilities.

Strategic Partners

Technology vendors (Microsoft, AWS, Salesforce) who refer clients needing implementation expertise

Referral Networks

Complementary service providers who recommend your firm for specialized capabilities

Direct Clients

Organizations seeking specific solutions through RFPs, industry connections, or reputation

These partnerships create a steady pipeline of qualified opportunities while allowing firms to focus on their core competencies.

Sales & Pre-Sales Team Role

1 **Opportunity Identification**

Monitoring partner portals, industry events, and client signals for project needs

2 **Initial Engagement**

Conducting discovery calls to understand client challenges, budget, and timeline

3 **Solution Design**

Pre-sales engineers create technical proposals aligned with client requirements

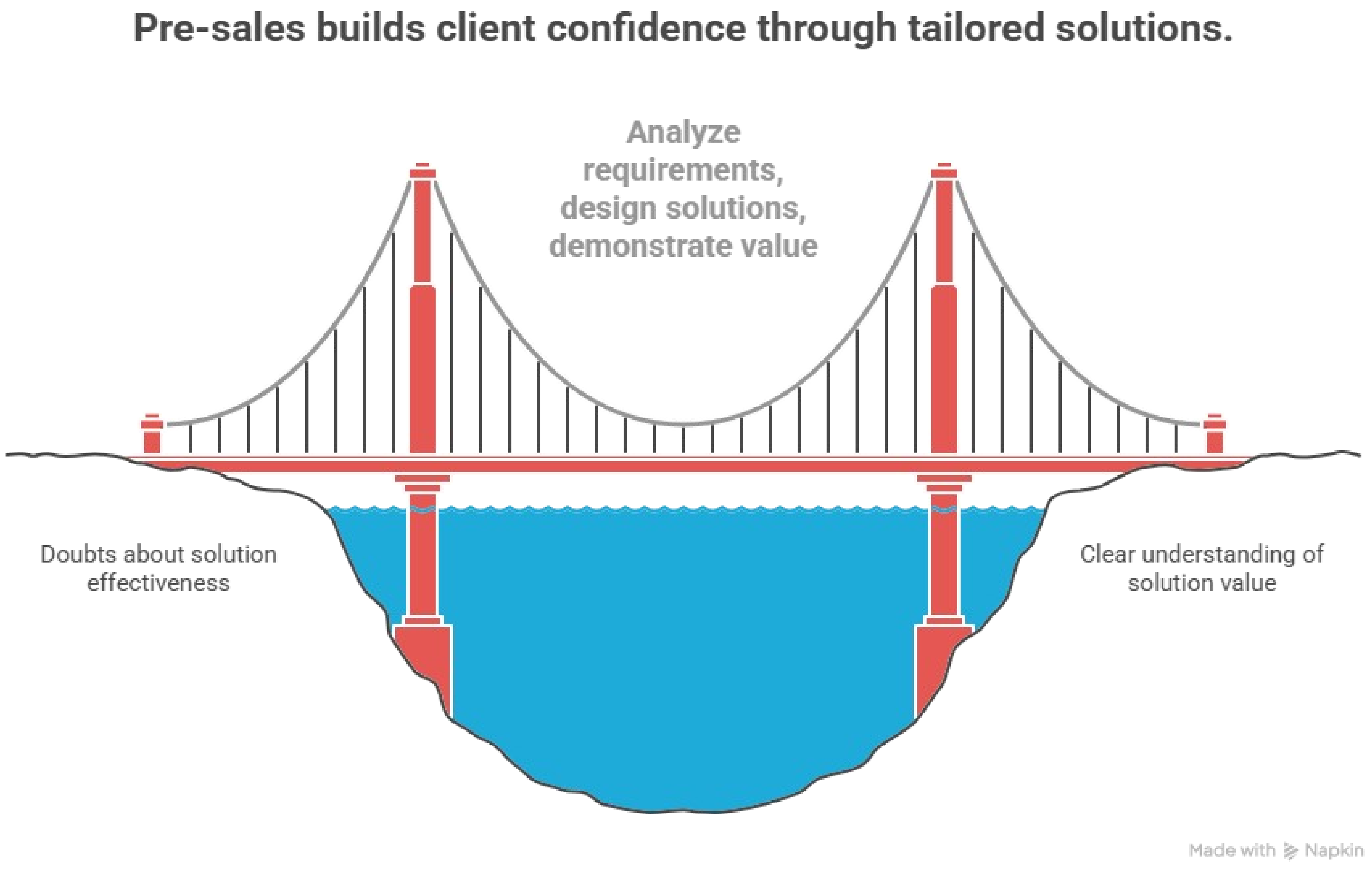
4 **Proposal Submission**

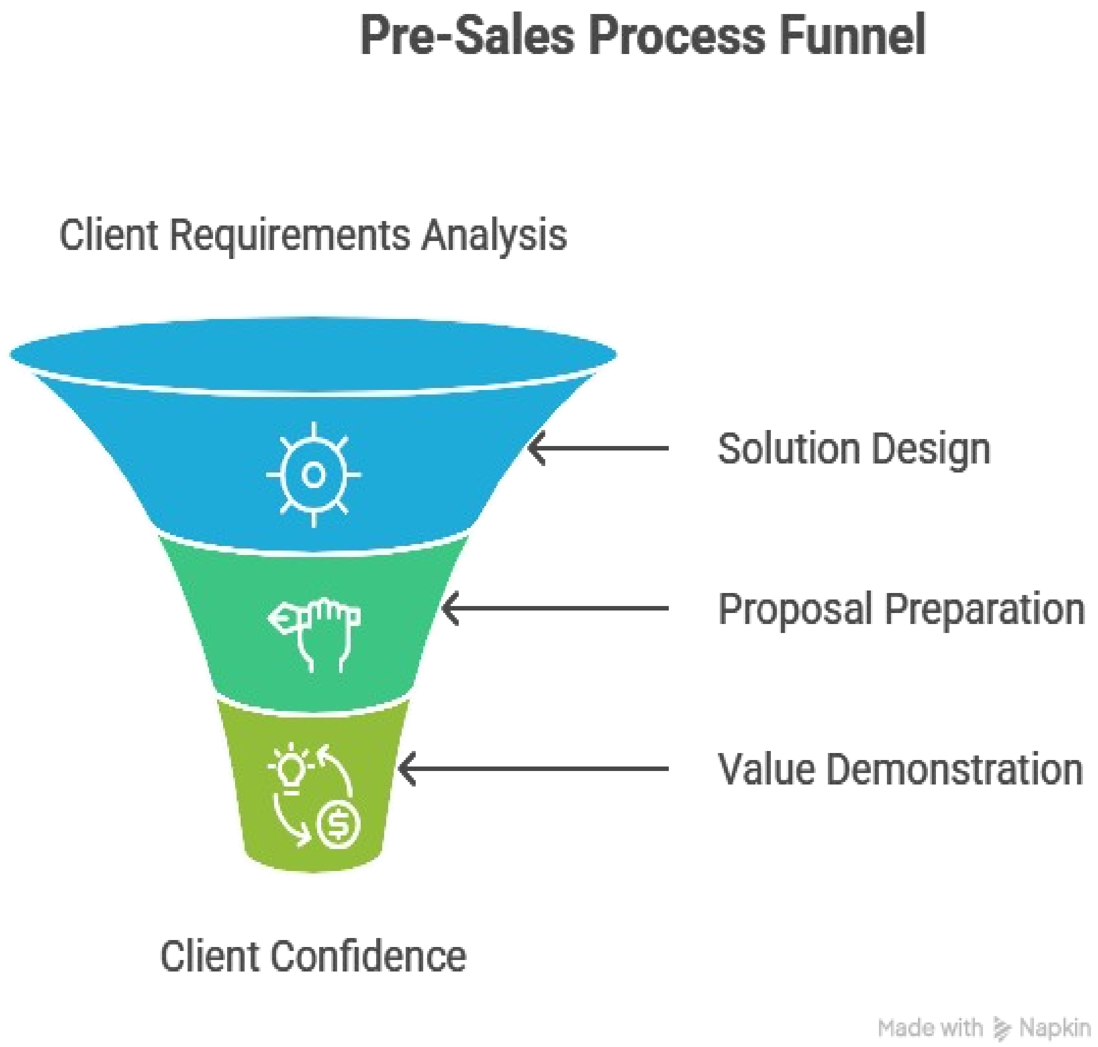
Delivering comprehensive proposals with scope, pricing, and success metrics

The Opportunity Hunt

Sales and pre-sales teams are the **frontline revenue drivers**, constantly identifying and nurturing potential projects. They monitor partner channels, respond to inbound inquiries, and proactively reach out to target accounts.

Sales & Pre-Sales Team Role





Key Insight: Pre-sales teams bridge the gap between sales promises and delivery reality. They ensure proposed solutions are technically feasible and properly scoped before contracts are signed.

Project Acquisition Process

Winning a project requires navigating formal procurement processes and demonstrating clear value. Companies typically acquire projects through three main channels:

1

Request for Proposal (RFP)

Formal bidding process where clients issue detailed requirements documents. Vendors submit comprehensive proposals including methodology, team qualifications, timeline, and pricing.

- Most common for government and large enterprise projects
- Highly competitive with multiple vendors competing
- Longer sales cycles (2-6 months typical)

2

Direct Quotation

Client requests pricing and approach for a specific, well-defined project. Less formal than RFPs, allowing for more consultative discussions.

- Common for existing clients or partner referrals
- Faster decision cycles (2-6 weeks)
- Relationship-driven rather than purely competitive

3

Strategic Partnership

Long-term agreements where the client pre-approves the vendor for specific types of work. Projects flow through master service agreements.

- Based on demonstrated past performance
- Simplified procurement for both parties
- Consistent revenue stream for the vendor

Proof of Concept (POC)

Before committing to a full project, clients often require a **Proof of Concept** — a small-scale demonstration that validates technical feasibility, solution quality, and vendor capability.

Typical POC Duration

2-4

Weeks

Average POC timeline

20%

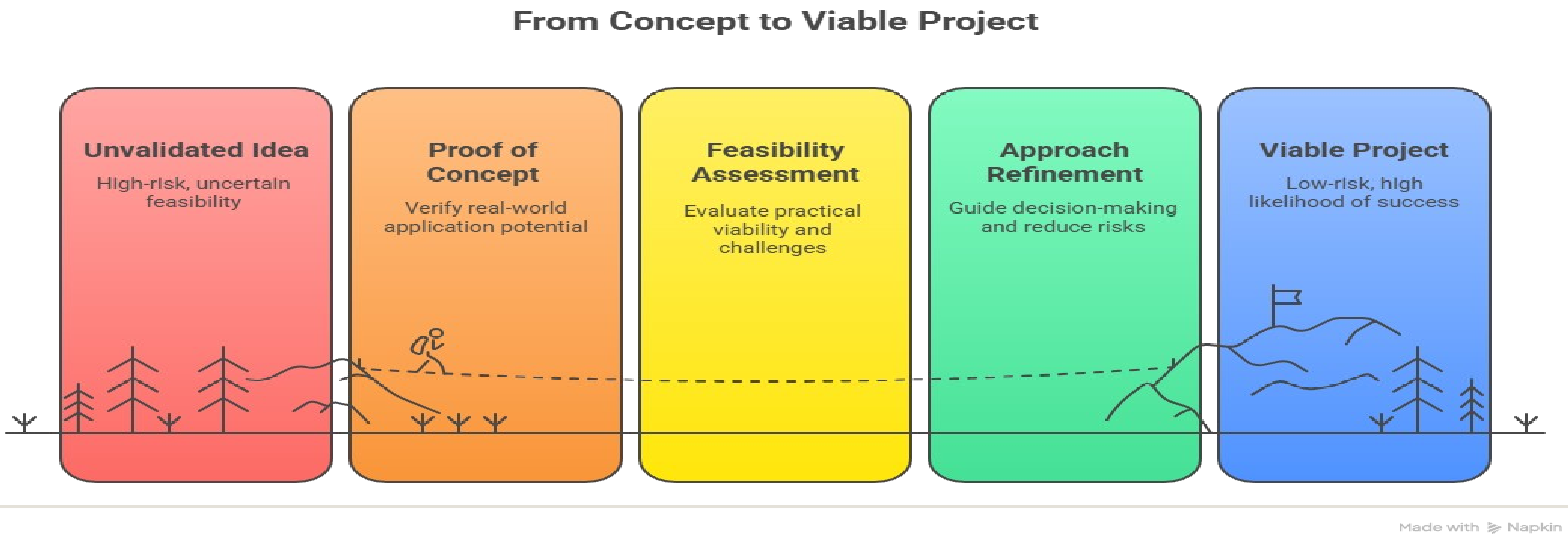
Of Full Scope

Typical POC coverage

POCs are often paid engagements but at reduced rates, demonstrating mutual commitment while minimizing client risk.

POC Objectives

- **Validate feasibility:** Confirm the proposed solution actually works with client systems and data
- **Assess capability:** Evaluate team skills, communication, and methodology
- **Reduce risk:** Identify potential challenges before major investment
- **Build confidence:** Create tangible evidence of value



Project Kick-Off Call

Once the contract is signed and the POC (if required) succeeds, both teams convene for a formal **kick-off meeting** — the official launch of the project partnership.

1

Introductions & Roles

All stakeholders introduce themselves, clarifying decision-makers, subject matter experts, and day-to-day contacts on both sides.

2

Project Objectives

Reconfirm business goals, success metrics, and expected outcomes to ensure complete alignment from the start.

3

Timeline & Milestones

Review the overall schedule, key deliverable dates, and any client-side dependencies or constraints.

4

Communication Protocol

Establish meeting cadence, reporting formats, escalation procedures, and collaboration tools.

5

Next Steps

Agree on immediate actions, typically scheduling the discovery workshop and requesting access to relevant systems.

Critical Success Factor: A well-run kick-off sets the tone for the entire engagement. Clear communication and aligned expectations at this stage prevent costly misunderstandings later.

Discovery Phase

The discovery phase is where the project team transitions from high-level proposals to **detailed, actionable requirements**. This investigative process ensures everyone understands exactly what needs to be built.

Stakeholder Interviews

One-on-one conversations with key users, managers, and decision-makers to understand their specific needs and pain points

System Analysis

Technical assessment of existing infrastructure, data sources, integrations, and any legacy systems that must be considered

Process Mapping

Documenting current workflows and designing future-state processes that the solution will enable or improve

Requirements Documentation

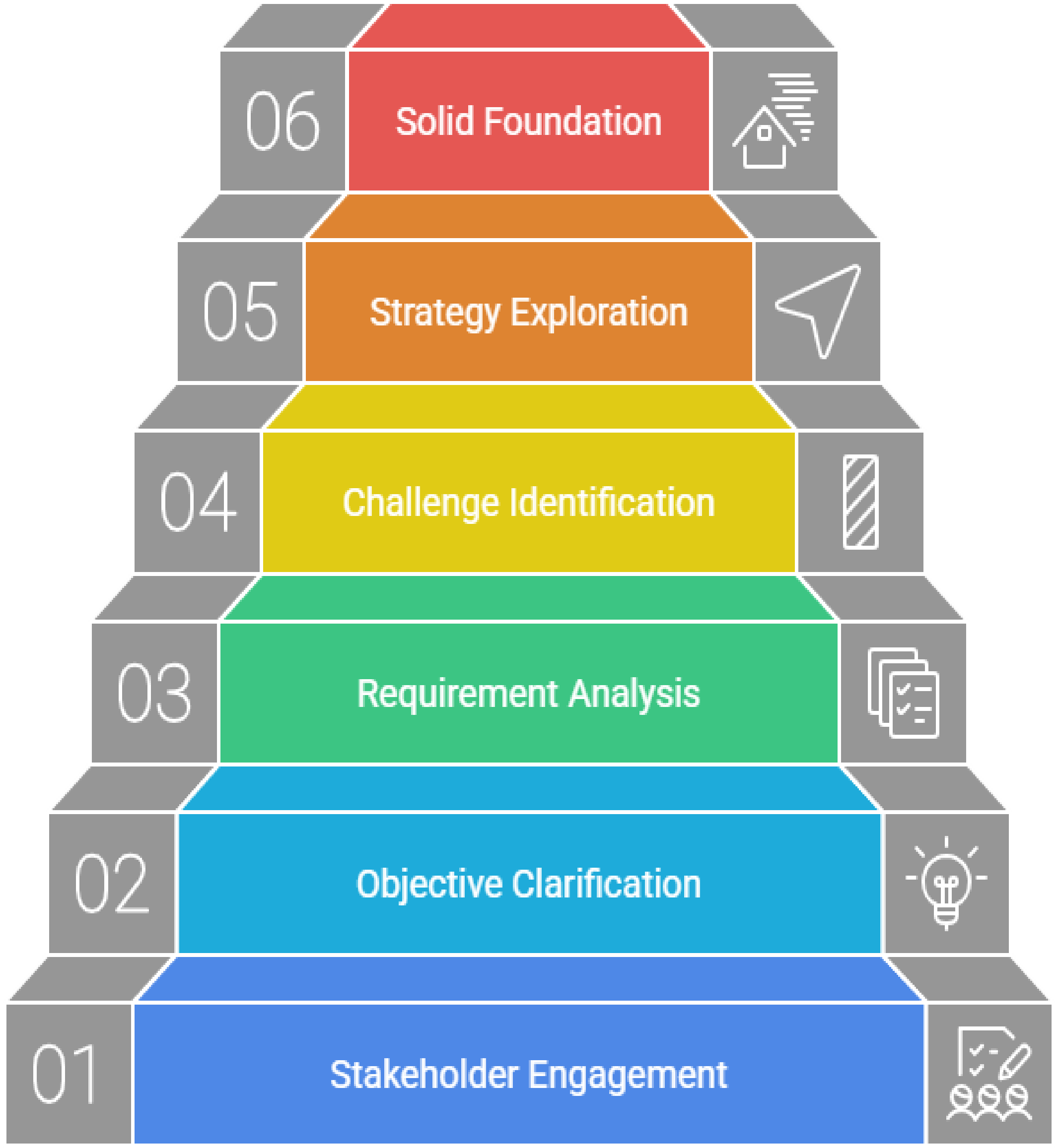
Creating comprehensive specifications that detail every feature, integration, and acceptance criterion

Discovery Deliverables

- Detailed requirements document
- Technical architecture diagram
- Data flow and integration specifications
- User stories and acceptance criteria
- Risk assessment and mitigation plan

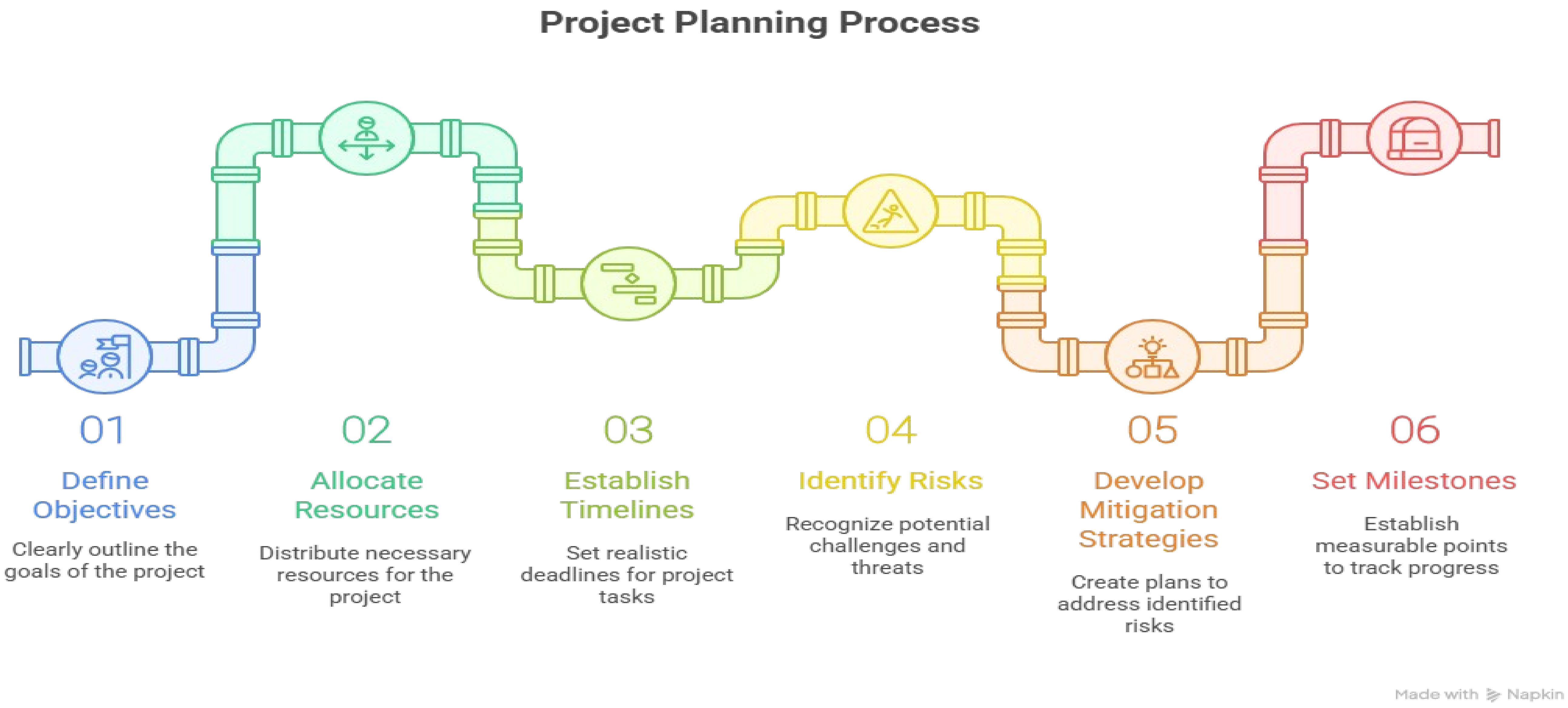
Common Discovery Activities

- Facilitated workshops with cross-functional teams
- System demonstrations and walkthroughs
- Data profiling and quality assessment
- Security and compliance reviews
- User experience design sessions



Project Planning

With requirements clarified, the delivery team creates a comprehensive **project plan** — the roadmap that guides execution from start to finish.



Project Planning

Scope Definition — 1

Finalizing exactly what will and won't be included, with clear boundaries to prevent scope creep

Resource Allocation — 2

Assigning team members, estimating effort, and securing necessary tools and environments

Sprint Planning — 3

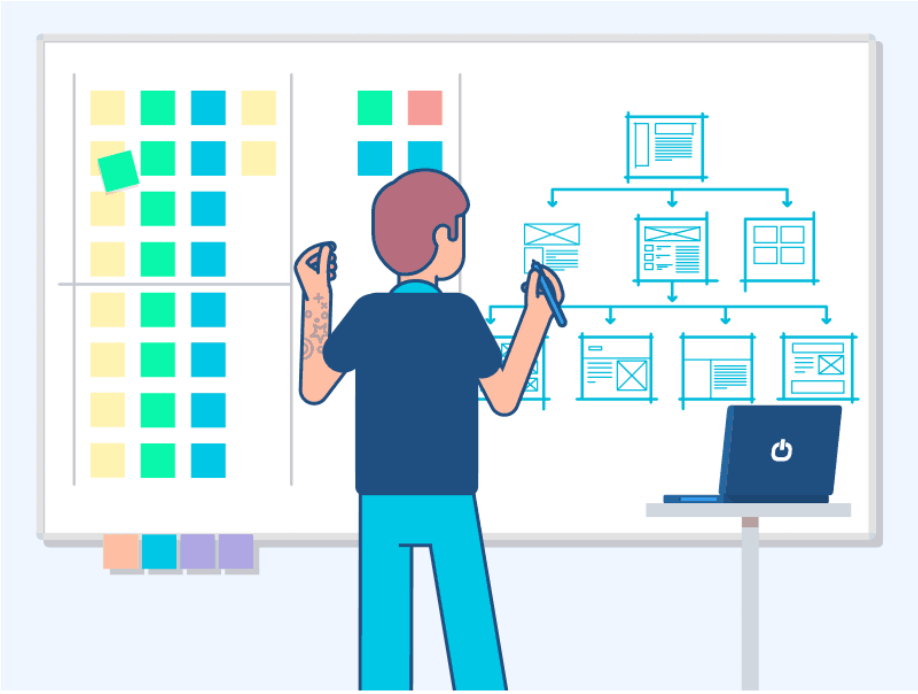
Breaking work into manageable sprints (typically 2-week iterations) with specific deliverables

Risk Management — 4

Identifying potential obstacles and creating contingency plans for likely scenarios

Quality Assurance — 5

Defining testing strategy, acceptance criteria, and validation procedures



Agile vs. Waterfall: Most modern service firms use **Agile methodologies** with iterative sprints, allowing flexibility as requirements evolve. However, some industries or project types still require traditional waterfall approaches with sequential phases.

Key Planning Documents

- Work breakdown structure
- Resource allocation matrix
- Communication plan
- Risk register
- Change control process

Typical Sprint Cycle

Sprint Planning

Team selects work items for the upcoming sprint

Development

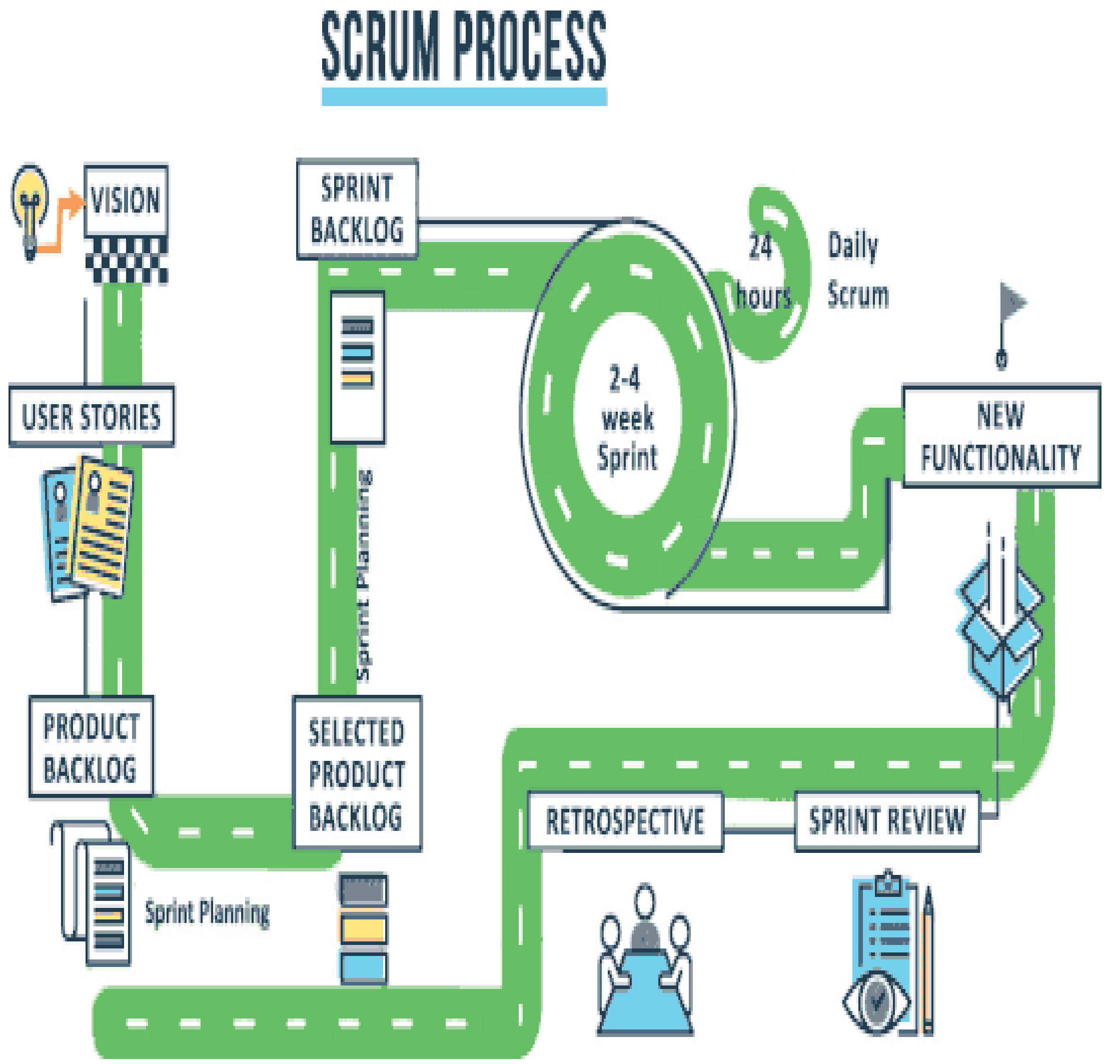
Engineers build and test features

Retrospective

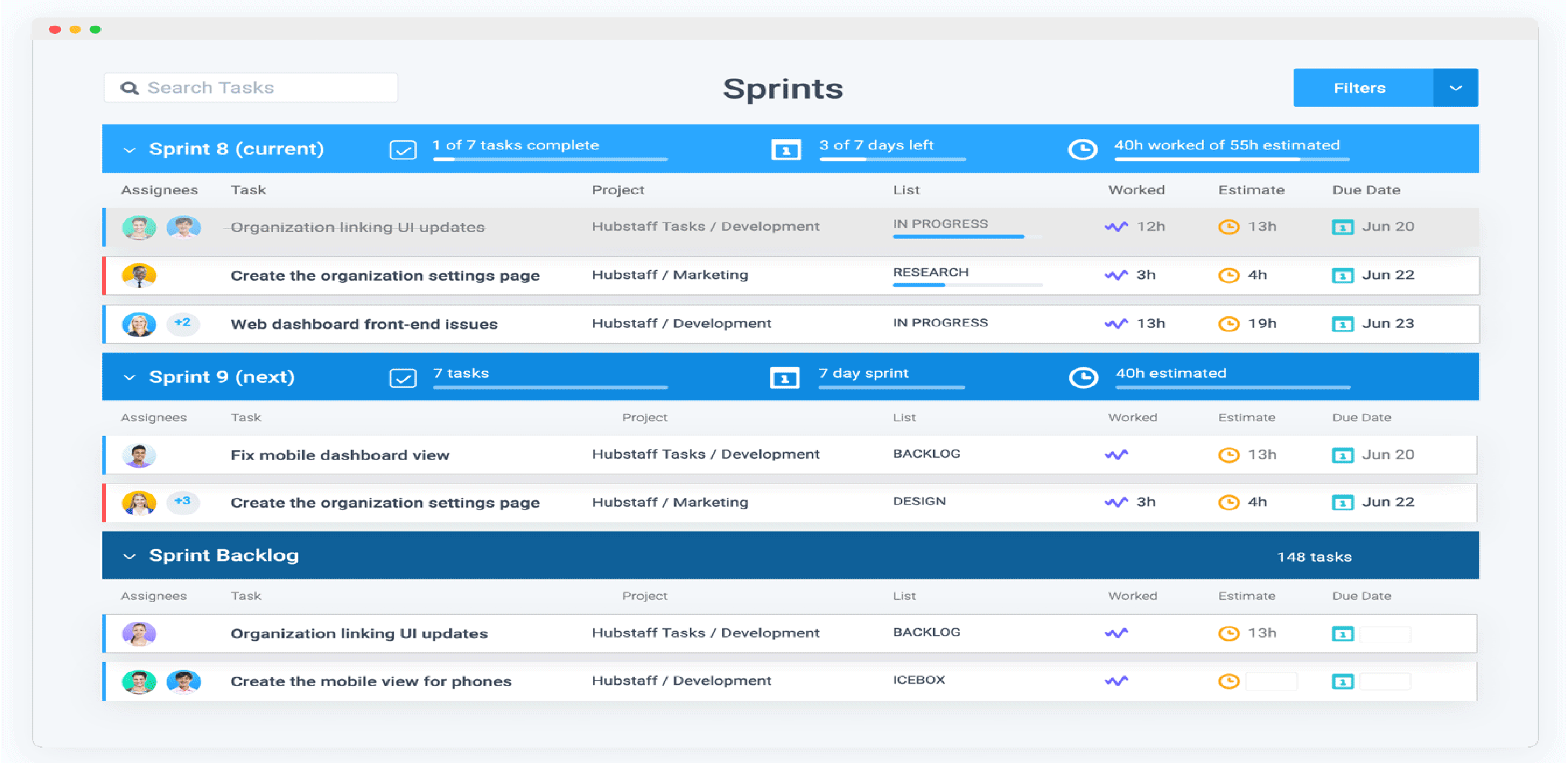
Team reflects and identifies improvements

Sprint Review

Demo completed work to stakeholders



Typical Sprint Cycle

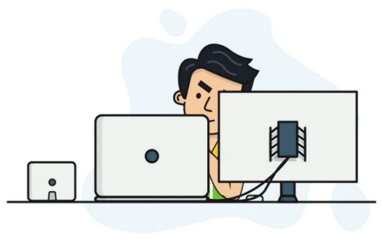


Execution Phase

This is where plans become reality. The delivery team implements the solution while maintaining continuous communication with stakeholders.

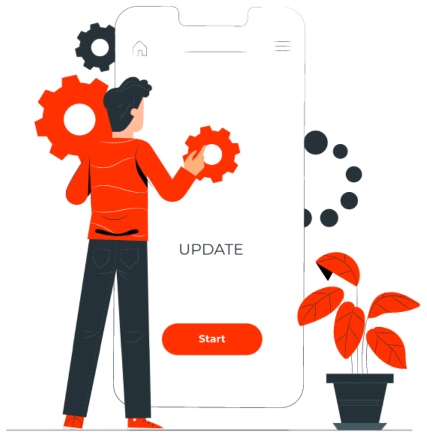
Development & Implementation

Engineers build features according to specifications, following coding standards and best practices. Work is organized into sprints with clear deliverables.



Quality Assurance

Continuous testing throughout development — unit tests, integration tests, user acceptance testing (UAT) — ensuring quality at every stage.

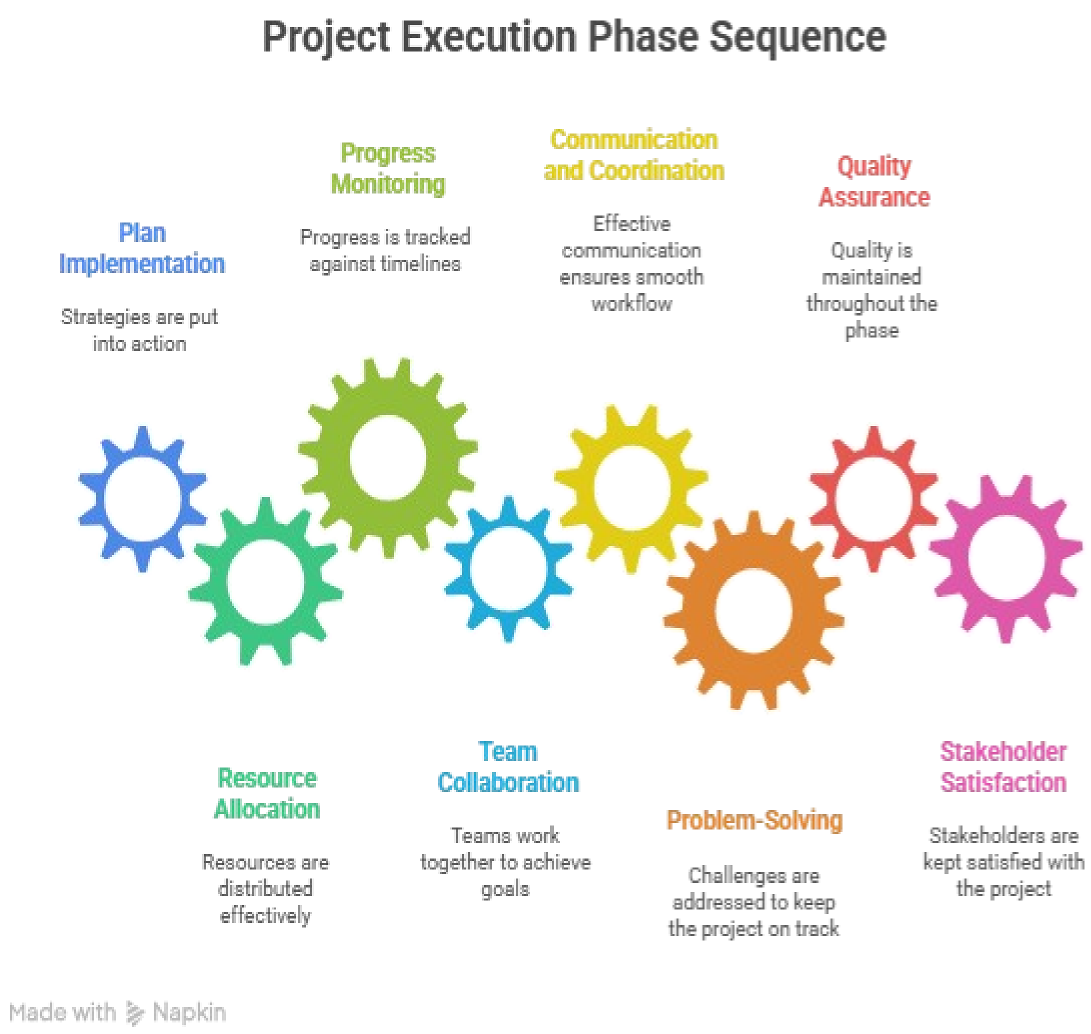


Regular Status Updates

Weekly or bi-weekly calls to review progress, demonstrate completed work, address blockers, and adjust priorities based on feedback.

Change Management

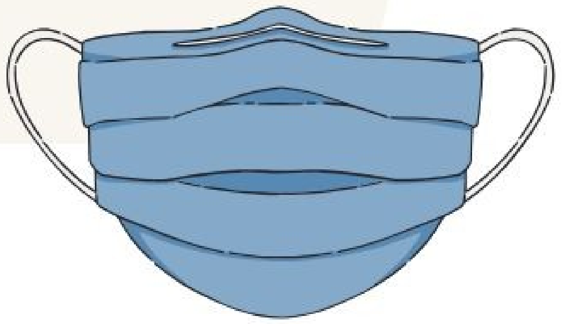
Formal process for handling scope changes, with impact analysis on timeline and budget before approval.



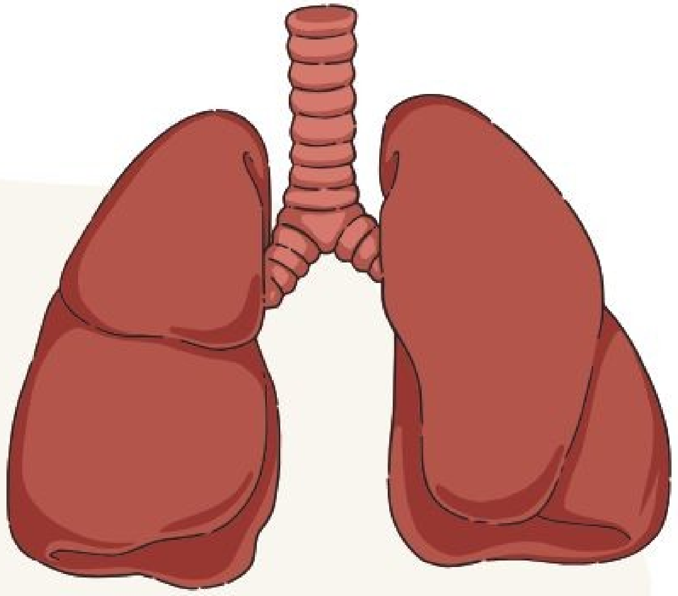
Live Use Case

Created By : Mahendra Singh

Axon Healthcare

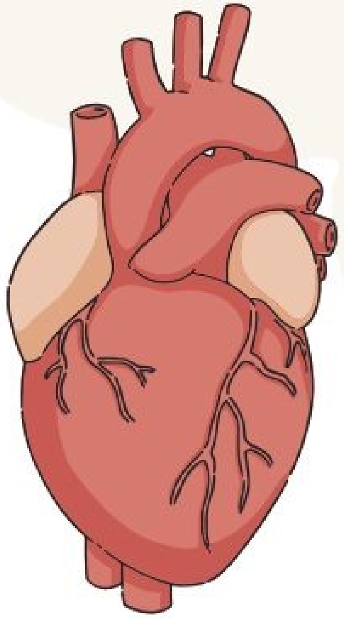


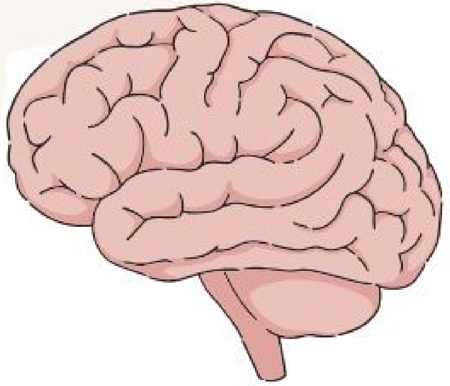
About Axon Healthcare



About Axon Healthcare - EMR Software Provider

Axon Healthcare is a health technology company dedicated to transforming how clinics, hospitals, and healthcare providers manage patient data. By offering secure, scalable, and user-friendly **Electronic Medical Records (EMR)** software, Axon Healthcare helps healthcare organizations streamline operations, enhance patient care, and ensure compliance with medical data regulations.





Problem Statement

For Business Intelligence project

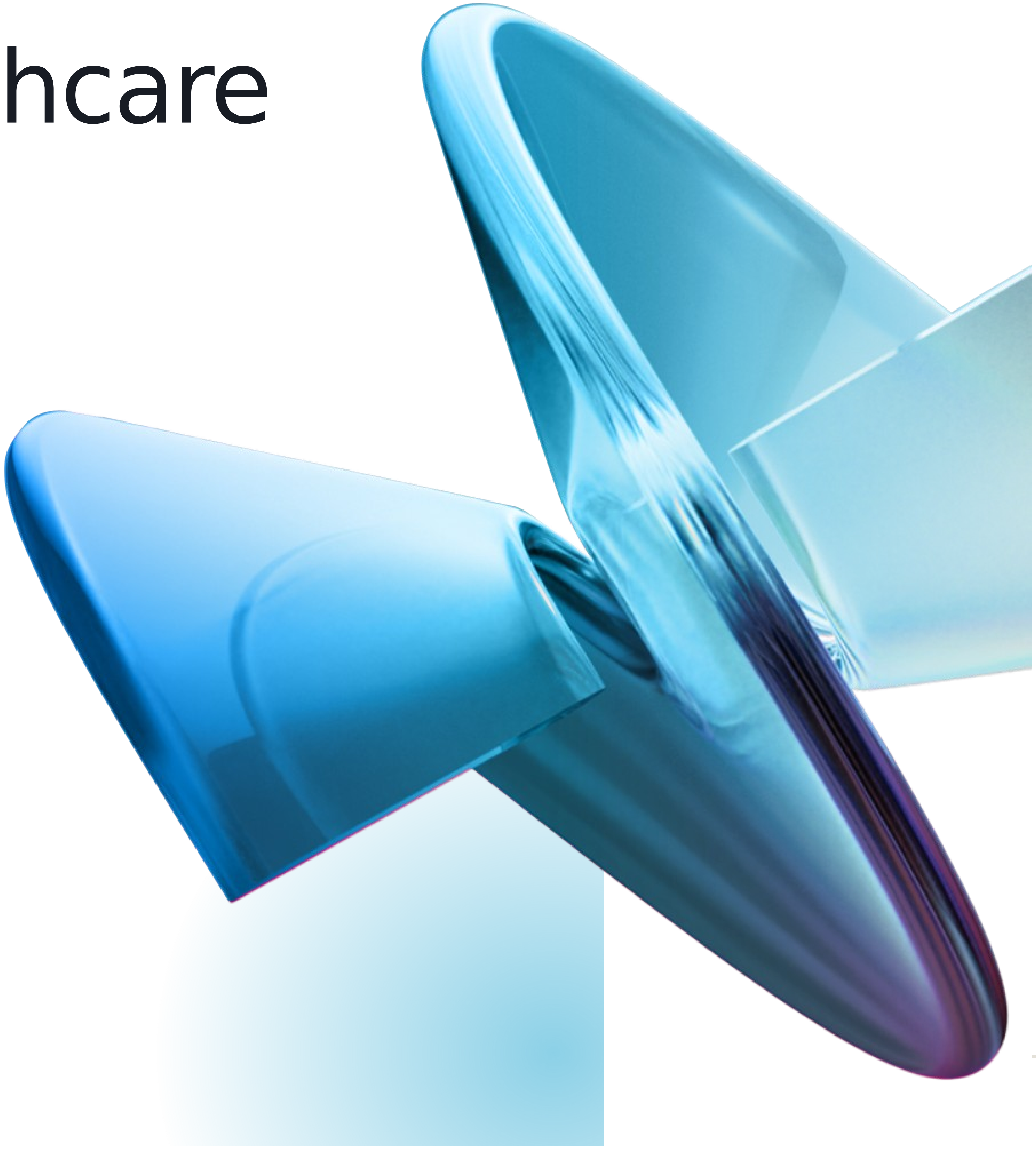
Axon Healthcare is struggling to convert EMR data into meaningful insights due to siloed systems, manual reporting, and lack of real-time visibility into clinical and operational KPIs. Stakeholders rely on outdated Excel reports, making decision-making slow and reactive.
To address this, Axon is hiring a Power BI consultant to:

- Integrate EMR, billing, and operational data
- Automate reporting processes
- Build interactive dashboards for real-time insights
- Improve performance tracking across clinics and departments

Project on Axon Healthcare Analytics

This project focuses on developing a BI dashboard for **Electronic Medical Records (EMR)** to provide actionable insights for healthcare providers and adn

Power BI



PROJECT LOGISTICS

LEAN AGILE METHODOLOGY

Meetings
- Team meeting/stand-up
- Delivery status meeting(s)
- Monthly Steering Meeting

Kickoff Meeting
- Team Introductions
- Services Overview
- Logistics
- Next Steps

Discovery Session
- Review current state
- Make best practice recommendations
- Technical onboarding
- Create backlog

Sprint Demo & Planning
- Completed sprint demo
- Retrospective
- Backlog grooming
- Next sprint planning

Project Closing
- Final review
- Knowledge transfer
- Next steps
- CSAT survey

Internal Team Creation

Team Collaboration Workflow

1. **BA** defines business and data requirements →
2. **Architect** designs technical solution →
3. **Data Engineer** builds backend pipelines →
4. **BI Developers** create and publish dashboards →
5. **PM** tracks progress, conducts reviews, and manages delivery

Kickoff Call With client

1. **Introductions & Roles** Team intros, key contacts, decision-makers

2. **Project Overview** EMR Analytics goals: improve patient flow, clinical KPIs, data visibility

3. **Scope** EMR → SQL → Power BI pipeline | Operational, Clinical, Financial & Regulatory dashboards

4. **Timeline** 10-week delivery: Discovery → Build → UAT → Go-Live

5. **Data & Access** EMR systems, SQL connectivity, security approvals

6. **Tech Stack** SQL Server | Power BI | ETL (SSIS/ADF) | On-prem/Azure

7. **Security & Compliance** HIPAA/GDPR, PHI handling, RLS, access controls

8. **Governance** Weekly status calls | Monthly reviews | Escalation path

9. **Risks & Mitigations** Data access delays, EMR dependencies, change management

10. **Success Metrics** 99% accuracy | <15 min refresh | 5+ dashboards | User adoption

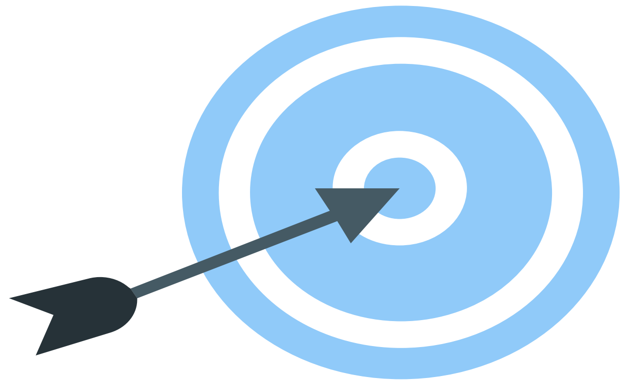
11. **Next Steps** Meeting minutes | Access setup | Discovery workshops | UAT scheduling

Discovery call Overview

To ensure all project requirements are gathered, verified and documented. To avoid any blind spots, scope creep and ensure project success.

High Level Design Presentation

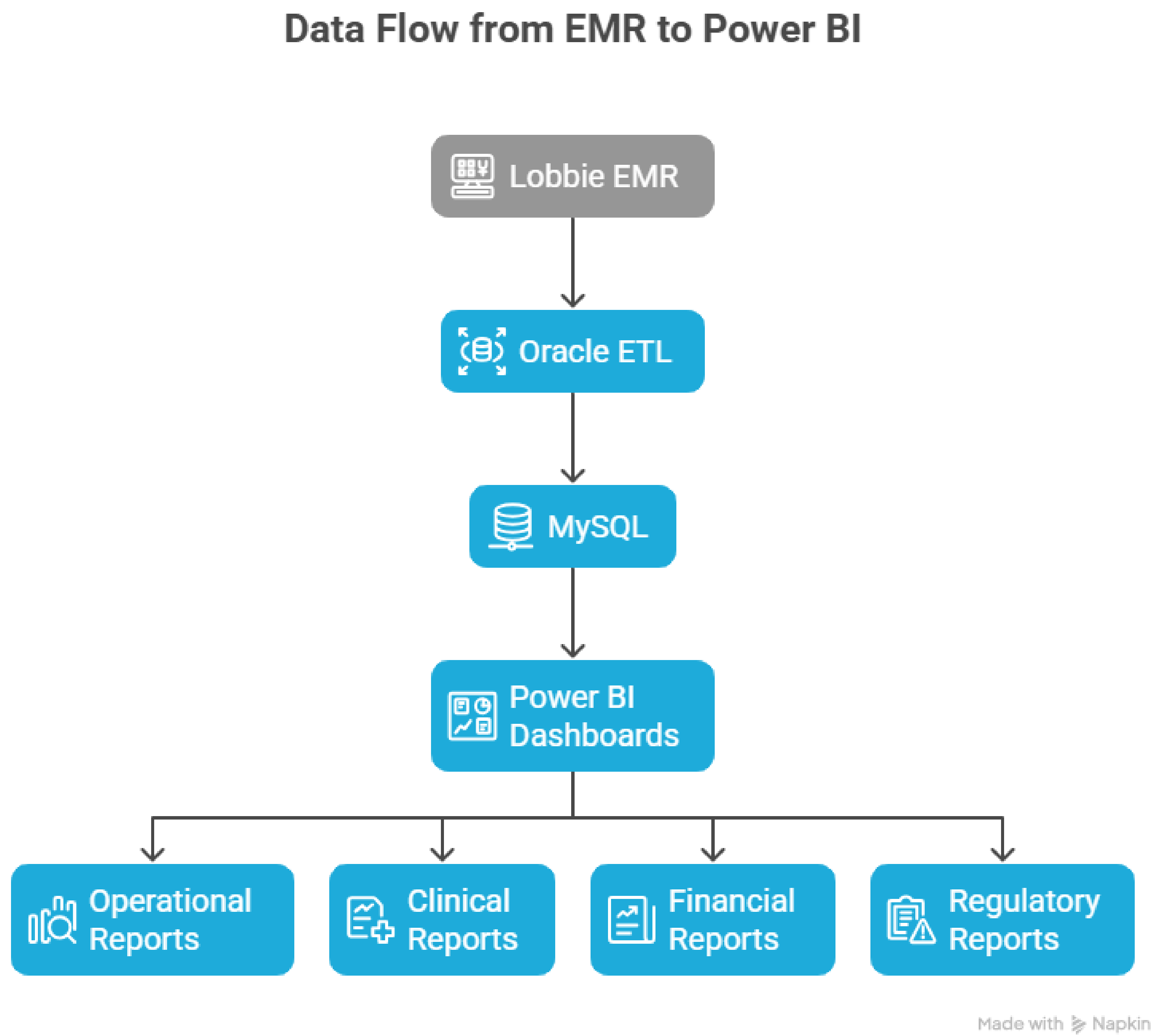
- Technical Architecture.
- High Level System Design.
- Project scope, assumptions & prerequisites.
- Execution Strategy
- New Data Warehouse Strategy
- Testing and Validation Approach
- Governance Structure



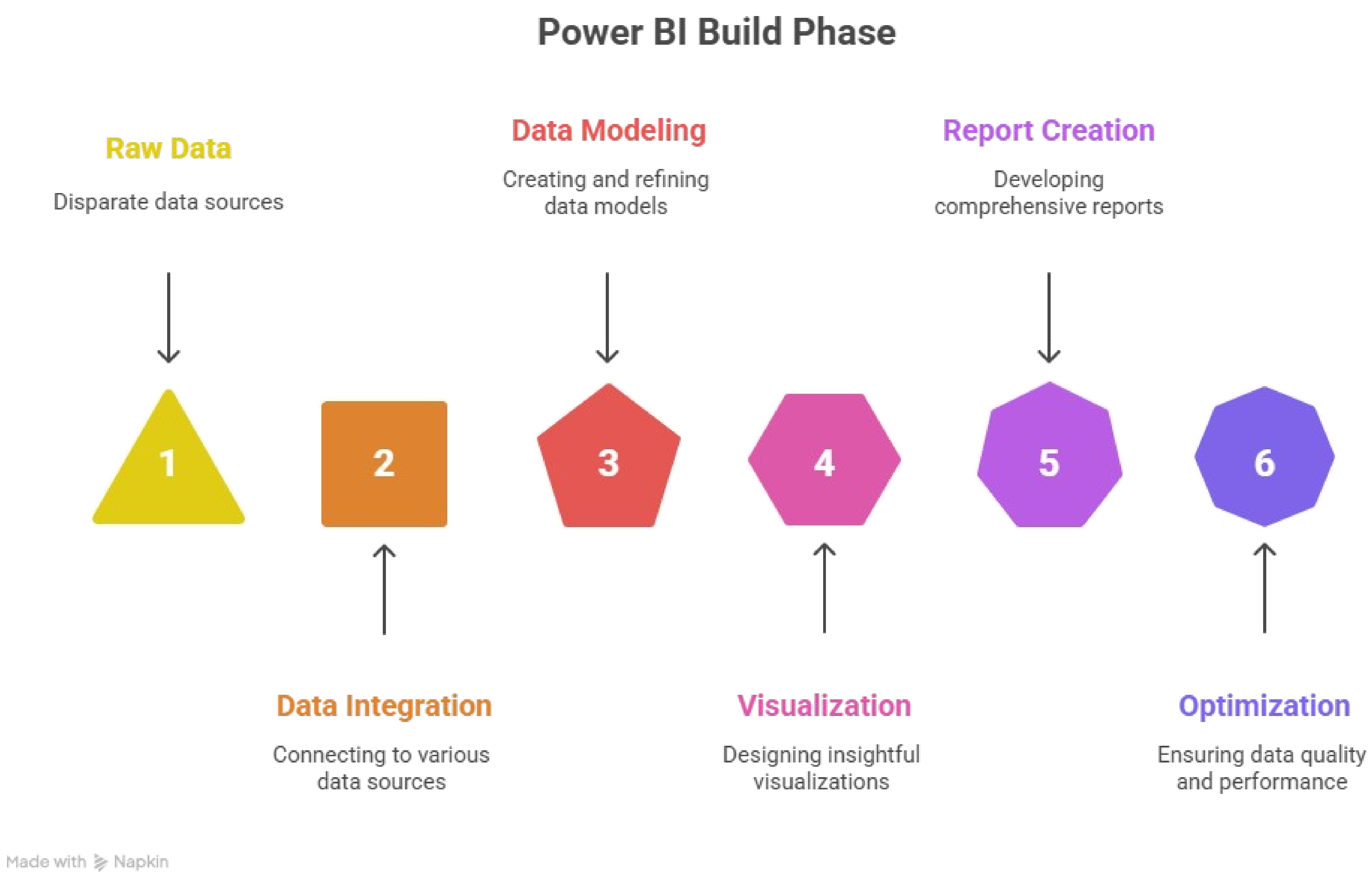
Project Plan

- High level execution plan.
- Work Breakdown Structure.
- Timeline and dependencies

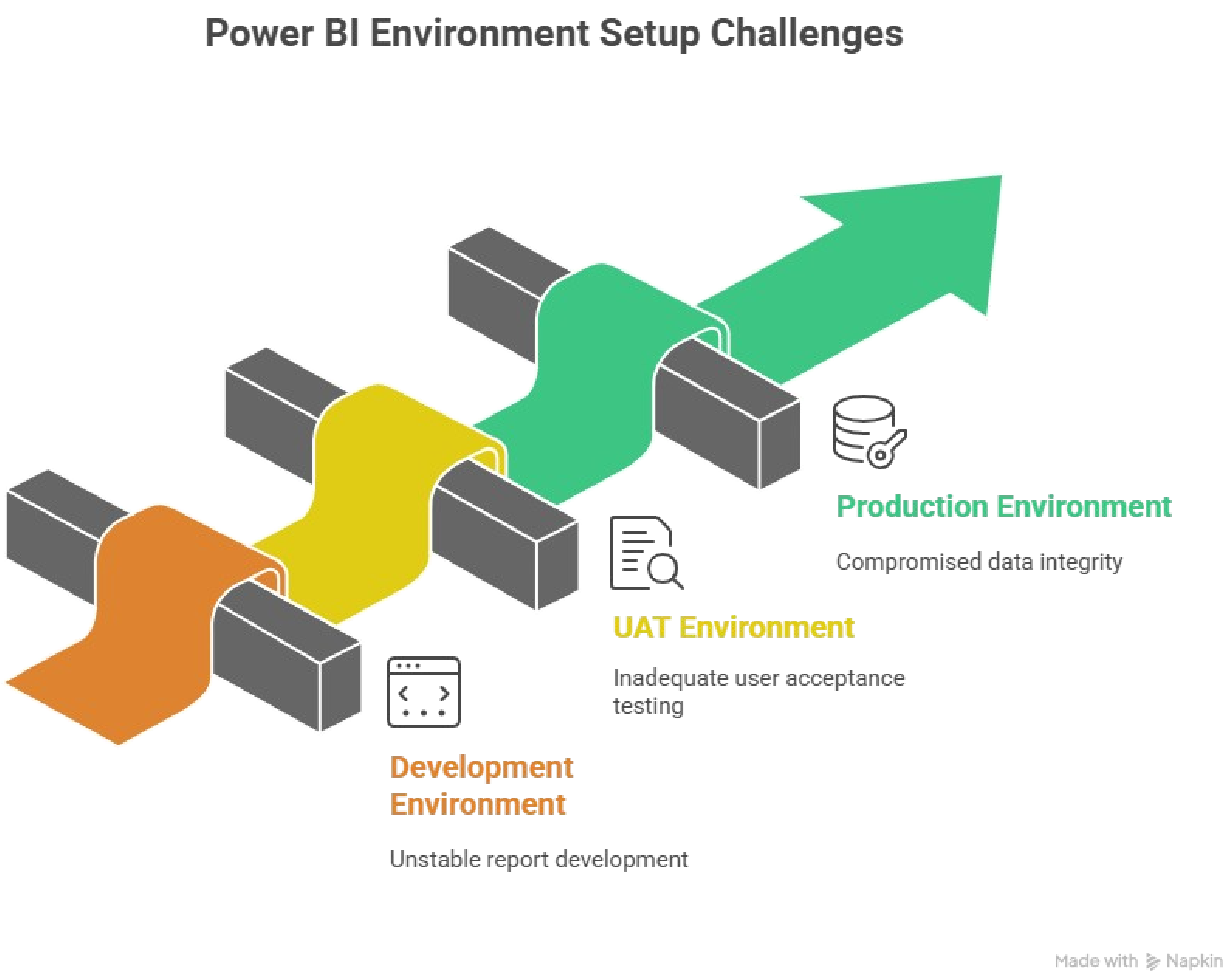
ETL Build Phase

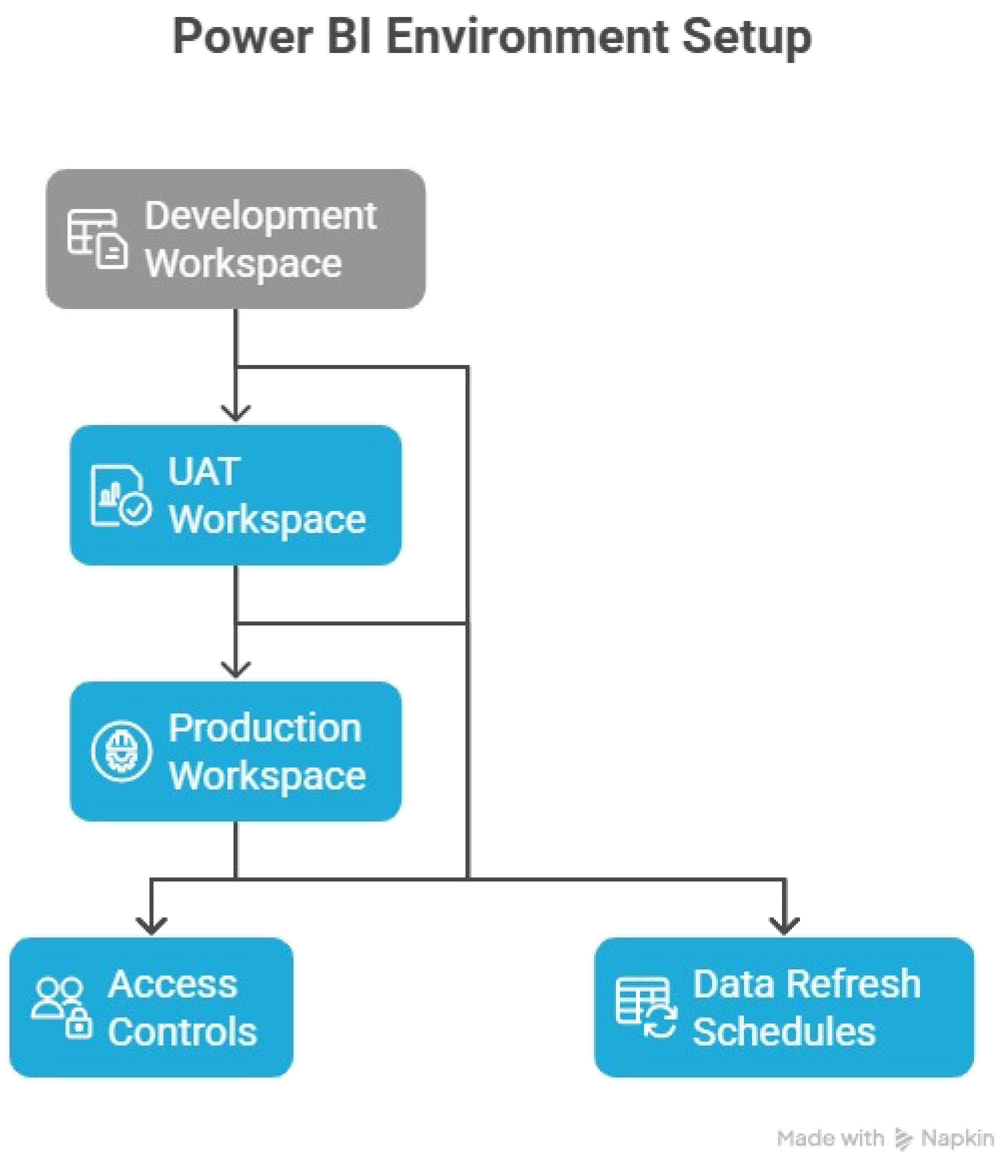


BI Build Phase



Environment Setup for BI





Data Modeling

1. Patients

patient_id	mrn	first_name	last_name	dob	gender	contact_info
1	MRN001	John	Smith	1985-03-15	Male	john.smith@email.com
2	MRN002	Sarah	Johnson	1992-07-22	Female	sarah.j@email.com

2. Patient_Addresses

address_id	patient_id	address_type	street	city	state	zip
1	1	Home	123 Main St	New York	NY	10001
2	2	Home	456 Oak Ave	Boston	MA	02101

3. Patient_Insurance

insurance_id	patient_id	provider_name	policy_number	effective_date
1	1	Blue Cross	BC123456	2024-01-01
2	2	Aetna	AE789012	2024-02-15

4. Encounters

encounter_id	patient_id	provider_id	admission_date	discharge_date	encounter_type
1	1	101	2024-10-01	2024-10-05	Inpatient
2	2	102	2024-10-15	2024-10-15	Outpatient

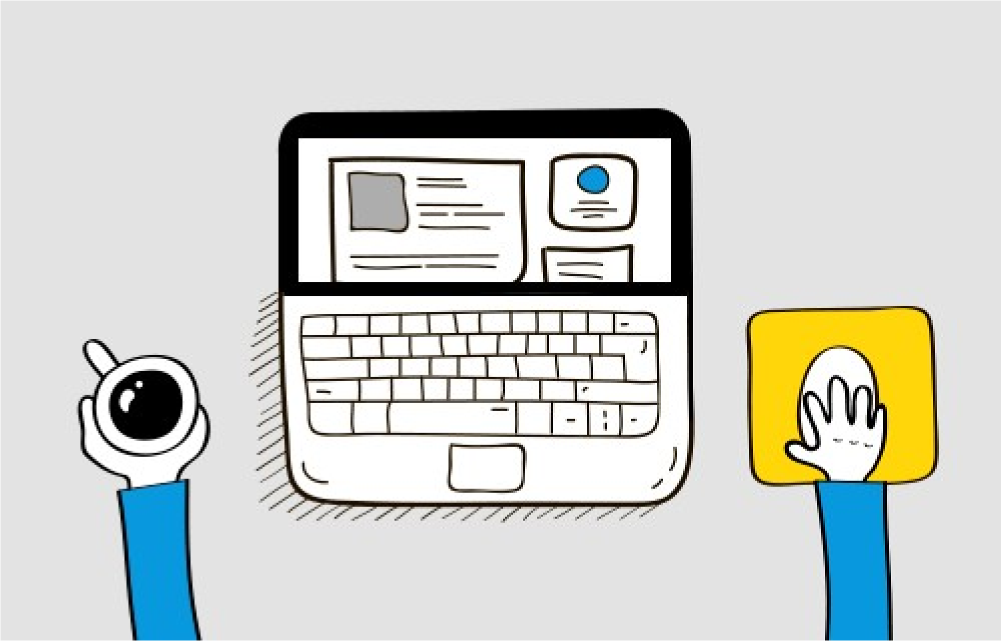
```
Patients (1) ──────────< (M) Encounters
    │                 │
    │                 ├───< Diagnoses
    │                 ├───< Procedures
    │                 ├───< Vitals
    │                 ├───< Lab Orders ──< Lab Results
    │                 ├───< Prescriptions ──< Medication Administration
    │                 └───< Billing ──< Charges
    │                          └───< Payments
    ├───< Patient Addresses
    ├───< Patient Insurance
    └───< Appointments

Providers (1) ──< (M) Encounters
    │
    └───── Belongs to ──> Departments

Medications (1) ──< (M) Prescriptions
```

KPI LIST

1. Total Patients
2. Total Doctors
3. Total Visits
4. Average Age of Patients
5. Top 5 Diagnosed Conditions
6. Follow-Up Rate
7. Treatment Cost Per Visit (Avg.)
8. Total Lab Tests Conducted
9. Percentage of Abnormal Lab Results
10. Doctor Workload (Avg Patients Per Doctor)
11. Total Revenue - Sum(Treatment Cost) + Sum(Visit Charges)

Dashboard Build Phase

1. **Development Process**
2. Dashboard development based on **BRD KPIs and standards**
3. **User stories** created for each requirement
4. **Daily progress reviews** and work tracking
5. **Client touchpoints** for logic clarification and business rule queries
6. **Iterative feedback loop** - build, review, refine
7. **Quality checks** on data accuracy and KPI calculations
8. **Progressive sign-offs** on completed dashboards



Dashboard UAT Phase

🧪 **UAT Process**

1. UAT preparation - test cases and scenarios ready

2. Test environment setup - replicate production with test data

3. User training sessions - brief end users on dashboard functionality

4. Test case execution by client/business users

5. Bug/issue logging - track defects and functional gaps

6. Daily defect review and prioritization (Critical/High/Medium/Low)

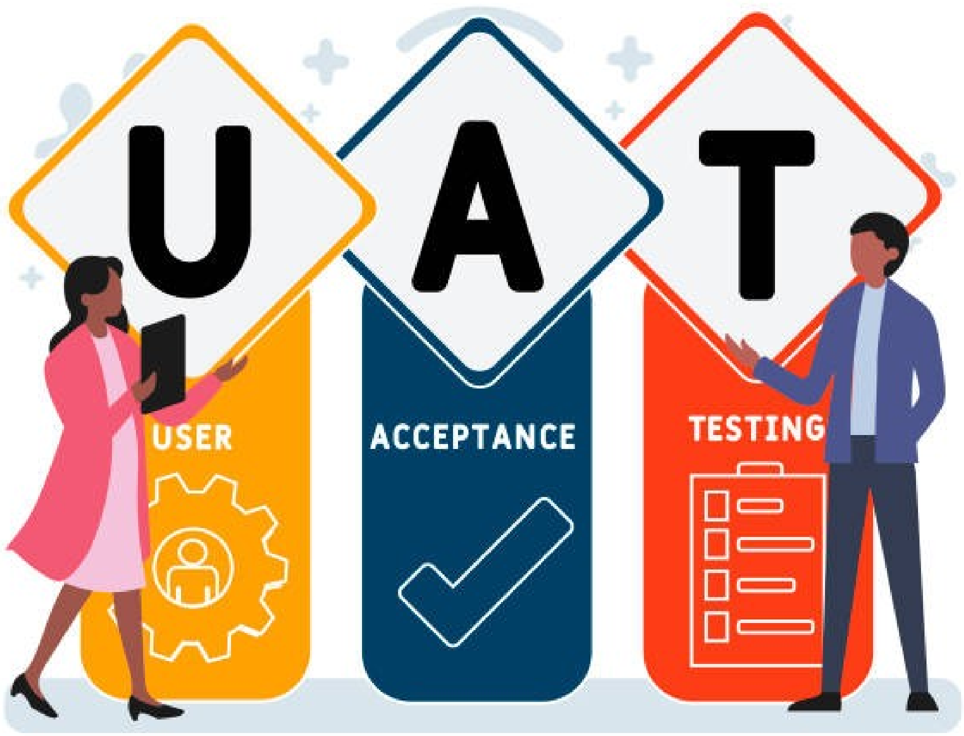
7. Fix and retest cycle - resolve issues and validate fixes

8. Data accuracy validation against source systems

9. Performance testing - refresh time and response speed

10. UAT sign-off - formal approval from client/stakeholders

11. Go-Live readiness review and deployment planning



Dashboard Prod Phase

Deployment Process

1. UAT sign-off received - formal approval from client

2. Production workspace setup in Power BI Service

3. Publish dashboards from Power BI Desktop to Prod workspace

4. Data gateway configuration - establish connection to MySQL database

5. Schedule data refresh - set automated refresh timings

6. Row-Level Security (RLS) implementation and testing

7. User access provisioning - assign roles and permissions

8. Share dashboards/app with end users

9. Go-Live announcement to all stakeholders

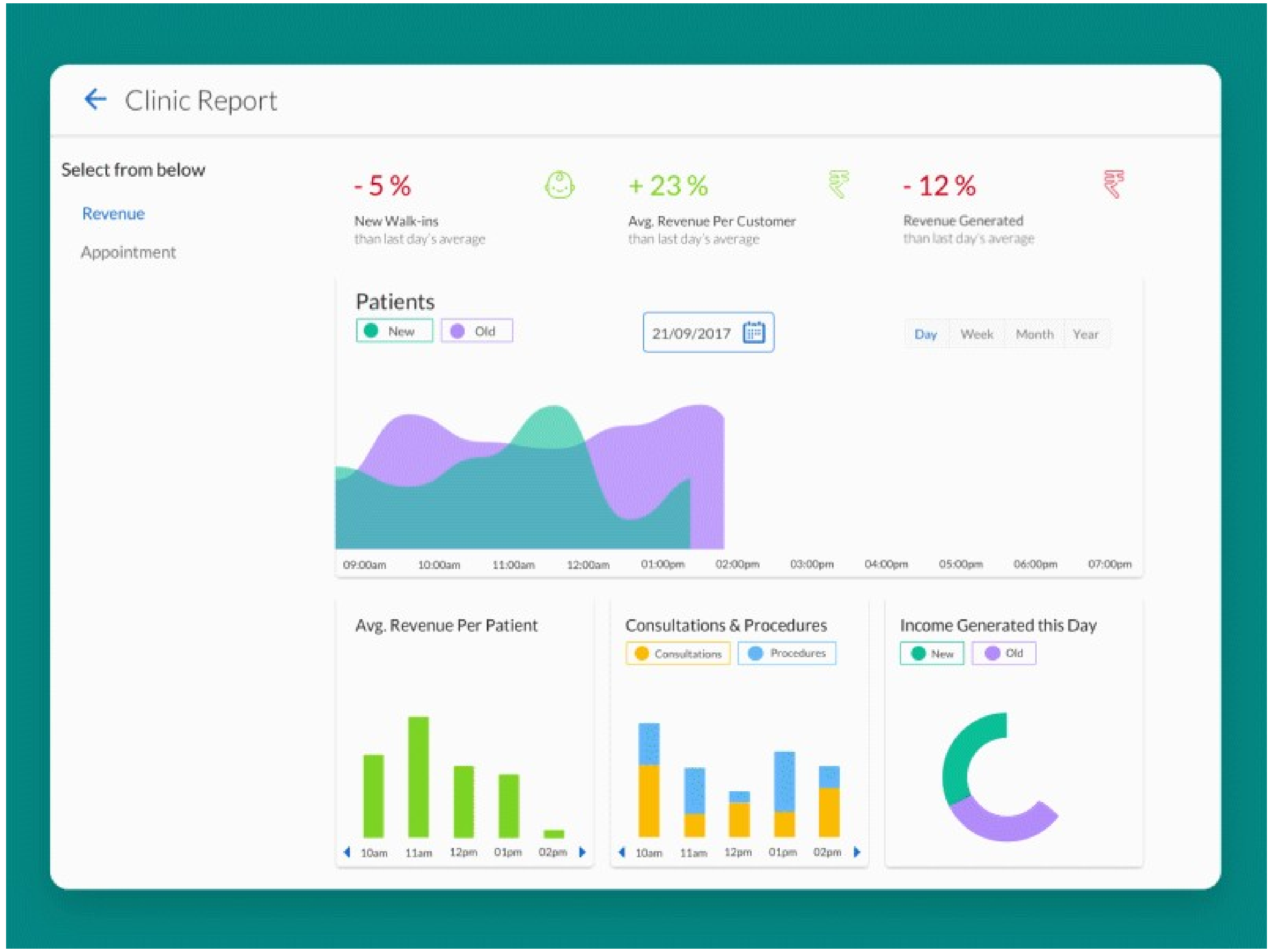
10. User training sessions conducted

11. Hypercare support - dedicated support for initial 2-4 weeks

12. Monitor dashboard performance - track refresh, usage, errors

13. Collect user feedback and log enhancement requests

14. Handover to support team with complete documentation



Dashboard


US Healthcare
Analysis Dashboard
Healthcare Overview
Hospital Insights
Patient Record and Prediction
Pherzy John Diez
Total Patients
55.50K
PY:
46K
YoY %:
20.29%
Total Billing Amount
$1.42bn
PY:
$1.18bn
YoY %:
20.37%
Avg Length of Stay
15.51
PY:
15.49
YoY %:
0.09%
Senior Patient %
38.10%
PY:
38.19%
YoY %:
0.22%
Emergency Rate %
32.92%
PY:
32.87%
YoY %:
0.14%
Abnormal Test Rate %
55.00%
PY:
55.07%
YoY %:
0.13%
Total Patients by Gender, Blood Type and Medical Condition
Non-binary 10.00%
Female 50.00%
Male 40.00%
Total Patients by Insurance Provider, Medical Condition and Hospital
Cigna 10.00%
Aetna 10.00%
Medicare 50.00%
dHealthCare 30.00%
Total Patients by Admission Type and Hospital
Emergency 32.92%
Elective 33.61%
Urgent 33.47%
Total Patients and Total Patients by Year, Quarter, Month and Day
7K
11K
11K
11K
11K
4K
2019
2020
2021
2022
2023
2024
Hospitals Clustered Map (KPI-centric)
12K
20K
16K
© US shape files by US Census Bureau (modified)
Go to Hospital Geo Insights for more details
Total Patients by Age Group, Blood Type and Medical Condition
Back
10K
12K
12K
12K
8K
0-18
19-30
31-45
46-60
61-75
76+
Total Patients by Doctor, Patient ID and Medical Condition
Michael Smith
27
John Smith
22

Change Request

Change Request Process

- **Change request raised** by client or stakeholders
- **Impact analysis** - assess effort, timeline, and dependencies
- **Change review meeting** with the client to discuss the scope and priority
- **Approval/rejection** by the project steering committee
- **Update project plan** - adjust timelines and resources if approved
- **Implement changes** in Power BI dashboards/data model
- **Testing & validation** of modified functionality
- **Client sign-off** on implemented changes
- **Documentation update** - BRD, technical specs, user guides

Change Request Type

📝 **Common Change Request Types**

New KPI/Metrics - Add or modify calculations

Visual changes - Chart types, colors, layout adjustments

Filter updates - Add/modify slicers and filter options

Data source changes - New tables, SQL query modifications

Drill-down features - Add navigation and hierarchical views

Performance optimization - Improve load time and refresh speed

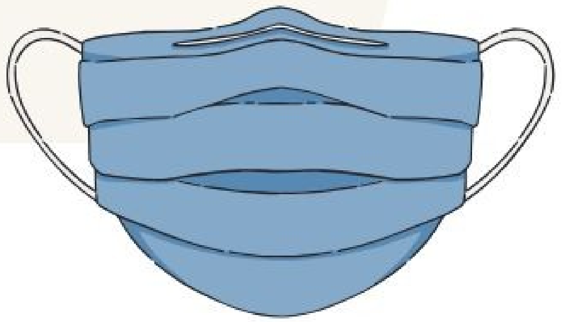
Access & security - RLS updates, user permission changes

Export options - Excel/PDF export, scheduled reports

Conditional formatting - Alerts, thresholds, color rules

UX improvements - Tooltips, mobile view, interactivity

Thank you for your attention



Questions!